

Assignment -2 CSCV

Section - A

Q.1 Encryption is the process of converting plain text into unreadable ciphertext to protect data from unauthorized access.

- Symmetric Encryption : Uses the same key for encryption and decryption (e.g., AES). It is fast but requires secure key sharing.
- Asymmetric Encryption : Uses two keys - a public key and a private key (e.g., RSA). It is more secure for communication but slower than symmetric encryption.

Q.2 Social engineering is a technique used by attackers to manipulate people into revealing confidential information.

E.g :- 1. Phishing : Attacker pretends to be a bank official to obtain account details.

2. Baiting : Offering free software downloads that contain malware.

Q.3 Phishing is a cyberattack where attackers send fraudulent emails pretending to be legitimate organizations to steal sensitive information.

Indicators of Phishing e-mails : 1. Suspicious sender email address.

2. Urgent language asking for immediate action.

Q.4 Malware is malicious software designed to damage, disrupt, or gain unauthorized access to system.

Types : 1. Virus : Attaches to files spreads when executed.

2. Ransomware : Encrypts victim's data and demands payment for decryption.

Q.5 A firewall is a network security system that monitors and controls incoming and outgoing network traffic based on security rules.

Types :- 1. Packet Filtering Firewall :- Filters traffic based on IP addresses and ports.

2. Stateful Inspection Firewall :- Monitors active connections and makes decisions based on traffic state.

Section - B

Q.1 Access control ensures that only authorized users can access specific resources, protecting confidentiality and integrity.

It helps :

- Prevent unauthorized access.
- Protect sensitive data.
- Enforce organizational policies.

Discretionary Access Control (DAC) : The owner of a resource decides who can access it.

E.g :- A file owner giving read permission to another user.

Mandatory Access Control (MAC):- Access based on security labels and classifications set by the system.

E.g :- Government systems where access depends on clearance level.

Difference : DAC is flexible and user-controlled, while MAC is strict and centrally controlled.

Q.2 Encryption key management refers to the secure generation, storage, distribution, rotation and destruction of cryptographic keys.

If keys are compromised, encrypted data become insecure.

Key Management Practices :

1. Secure Key Storage : Uses Hardware Security Modules (HSM) or encrypted storage to protect keys.
2. Key Rotation and Expiry : Regularly change encryption keys to reduce the risk of compromise.

Proper key management ensure data confidentiality and compliance with security standards.

A Denial of Service (DoS) attack attempts to make a system or network unavailable by overwhelming it with traffic.

Types of DoS Attacks :-

1. Flooding Attack :- Sends excessive traffic to exhaust bandwidth.
2. SYN Flood Attack :- Exploits TCP handshake process to consume server resources.
3. Application Layer Attack :- Targets web applications with repeated HTTP requests.

Mitigation :-

- Use firewalls and intrusion prevention systems.
- Rate limiting and traffic filtering.
- Deploy DDoS protection services.

Section - C

Q.1 Threat intelligence is the collection and analysis of information about current and emerging cyber threats.

Importance :-

- Helps identify potential risks.
- Enables proactive defence.
- Improve incident response.
- Reduces financial and reputational damage.

How Organizations use Threat Intelligence :-

- Updating firewall and IDS rules.
- Blocking malicious IP addresses.
- Strengthening security controls.
- Training employees about new attacks.

Sources of Threat Intelligence :-

- Government security advisories.
- Cybersecurity vendors.
- Information sharing platforms (ISACs)
- Open-source intelligence (OSINT)

Q.2 An Intrusion Detection System (IDS) monitors network or system activities to detect suspicious behavior or policy violations.

Role in Network Security :-

- Detects unauthorized access.
- Identifies malware activity.
- Generates alerts for administrators.
- Supports incident investigation.

Types of IDS :-

1. Host - Based IDS (HIDS) :- Installed on individual devices.
 - Monitors system logs and file changes.
 - Detects insider attacks.
 - Limited to one host.
2. Network - Based IDS (NIDS) :- Monitors network traffic.
 - Detects attacks across the network.
 - Covers multiple devices.
 - ~~can~~ Cannot inspect encrypted traffic easily.

Comparison :- HIDS provides detailed system - level monitoring, while NIDS provides broader network - level monitoring.
Using both together improves security.

Q.3 A cybersecurity policy is a formal document that defines how an organization protects its information systems.

Steps Involved :-

1. Identify organizational assets and risks.
2. Conduct risk assessment.
3. Define security objectives.
4. Establish roles and responsibilities.
5. Develop security controls and procedures.
6. Implement monitoring and enforcement mechanisms.
7. Review and updates the policy regularly.

Key Components :-

- Access control policy.
- Data protection and encryption guidelines.
- Incident response plan.
- Acceptable use policy.
- Password management policy.
- Backup and Recovery procedures.
- Compliance and legal requirements.

A comprehensive cybersecurity policy ensure systematic risk management and protects organizational data effectively.